



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2015
Higher 2

BIOLOGY

9648/01

Paper 1 Multiple Choice

Thursday, 17 September 2015

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

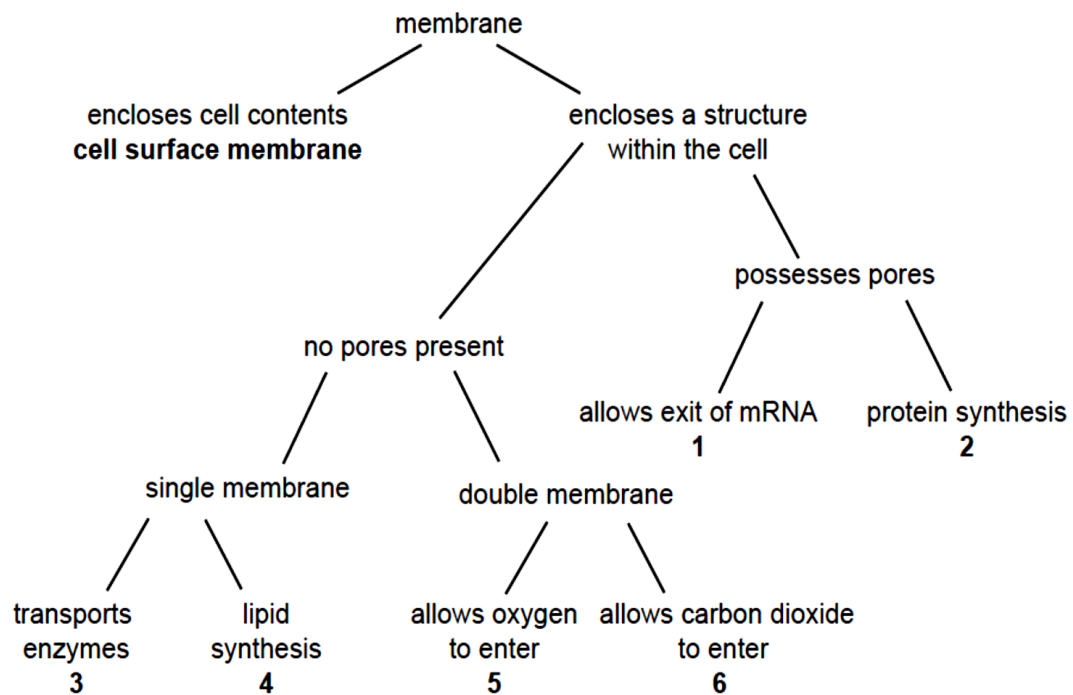
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

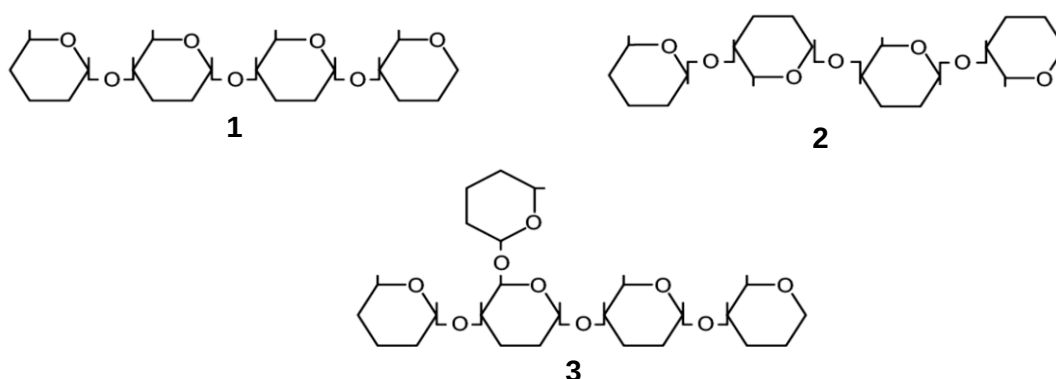
- 1 Membranes within and at the surface of cells have different roles. The flowchart shows a scheme to identify various membranes/ organelles within the cell, by describing the structure and function of the membranes/ organelles.



Which of the following outcomes correctly identifies the various membranes/ organelles?

	1	2	3	4	5	6
A	chloroplast	vesicle	smooth ER	rough ER	nucleolus	mitochondrion
B	nucleolus	rough ER	vesicle	smooth ER	nucleus	mitochondrion
C	nucleus	smooth ER	vesicle	rough ER	mitochondrion	chloroplast
D	nucleus	rough ER	vesicle	smooth ER	mitochondrion	chloroplast

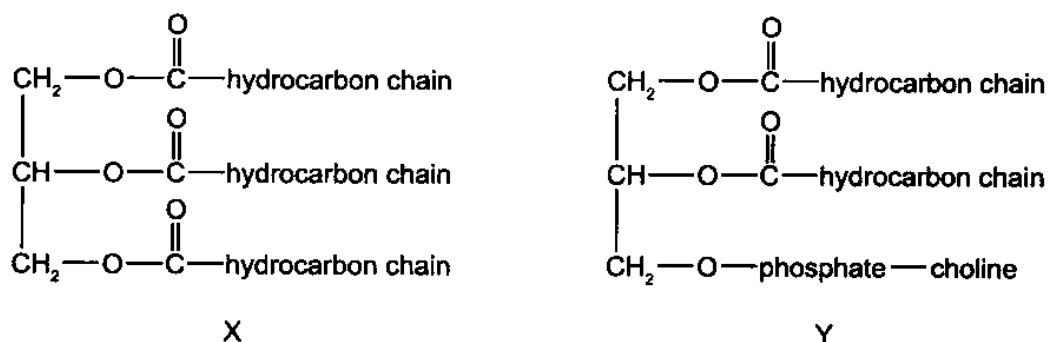
- 2 The figure below shows the structural formulae of three polysaccharides.



Which of the following is correct?

	1	2	3
A	amylose	cellulose	glycogen
B	amylose	amylopectin	cellulose
C	cellulose	amylose	amylopectin
D	glycogen	cellulose	amylopectin

- 3 The diagram shows the structure of two lipid molecules.

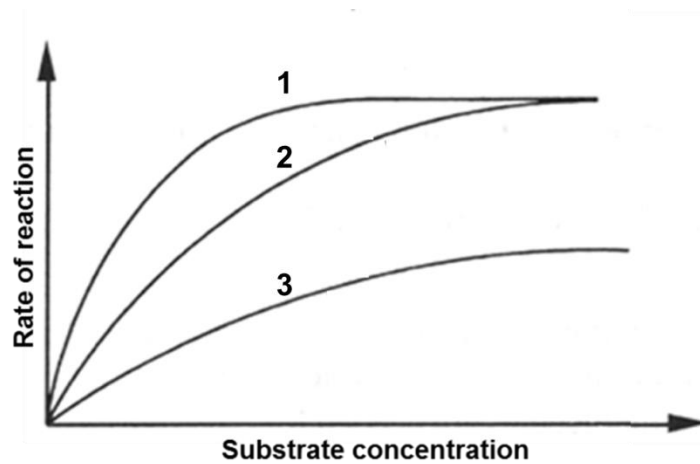


Which of the following statements correctly describe the two lipid molecules?

- Both molecules are esters of glycerol.
- Both molecules are made by condensation reactions.
- The hydrocarbon chains of molecule X and Y may be from saturated or unsaturated fatty acids.
- The hydrocarbon chains of molecule X are always the same length.

- A 1 and 2 only
 B 1, 2 and 3 only
 C 1, 3 and 4 only
 D All of the above

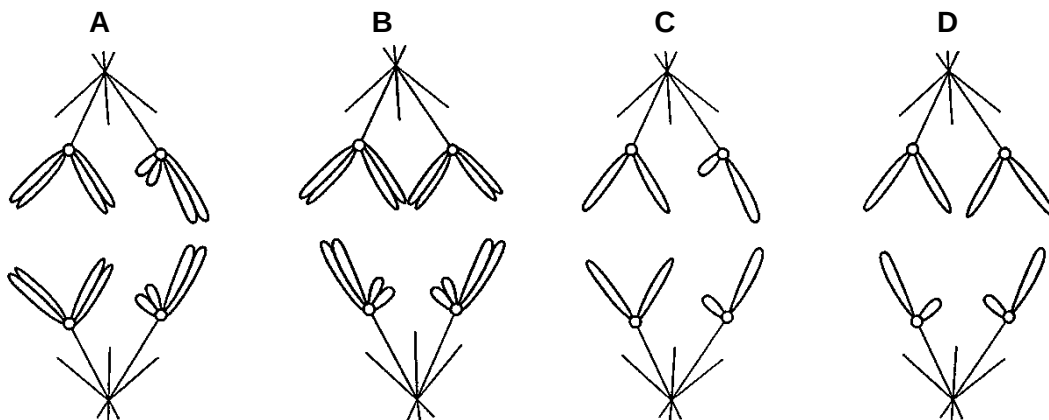
- 4 The curves show the rates of reaction of an enzyme with and without inhibitors.



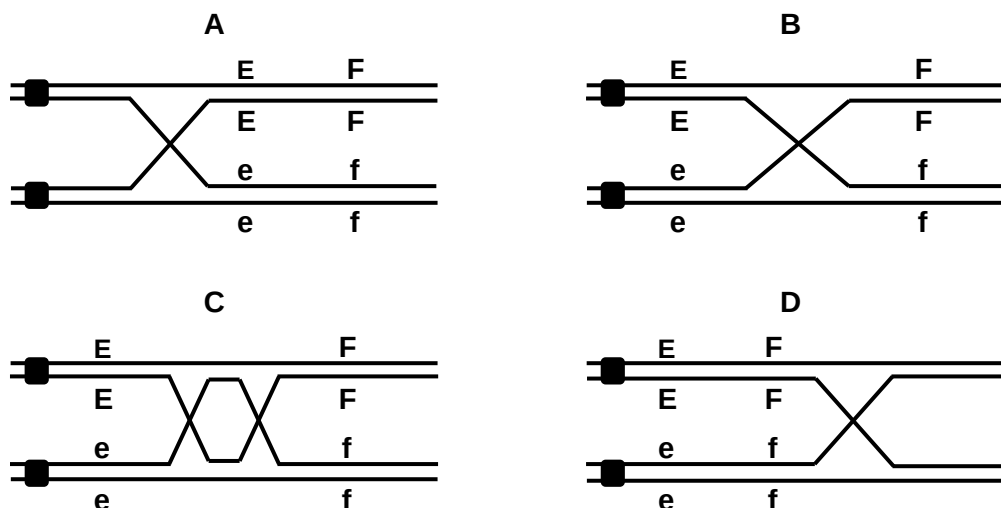
What do the curves show?

	curve 1	curve 2	curve 3
A	competitive inhibition	non-competitive inhibition	normal activity
B	competitive inhibition	normal activity	non-competitive inhibition
C	normal activity	competitive inhibition	non-competitive inhibition
D	normal activity	non-competitive inhibition	competitive inhibition

- 5 Which diagram shows anaphase I of meiosis of a nucleus which contains two pairs of homologous chromosomes?



- 6 The loci of two genes **E/e** and **F/f** are on the same chromosome. Which chromosomal event in meiosis increases genetic diversity in the next generation?



- 7 In 1958, an experiment was carried out to investigate the way in which DNA replicates. *Escherichia coli* bacteria were grown in a medium containing $^{15}\text{NH}_4\text{Cl}$. After very many generations all the bacterial DNA contained ^{15}N and was described as 'heavy'.

The bacteria were then transferred to a medium containing $^{14}\text{NH}_4\text{Cl}$. Some of the bacteria were removed as soon as they had reproduced once (first generation). Further samples were removed after the second and third generations.

The bacterial DNA was then extracted from each sample, placed in a solution of caesium chloride and spun in a centrifuge. The percentage of ^{15}N (heavy) DNA in each sample was then calculated.

What will be the percentage of cells containing ^{15}N in the third generation?

	Percentage of DNA in the sample obtained from the third generation/ %		
	$^{14}\text{N}-^{14}\text{N}$	$^{14}\text{N}-^{15}\text{N}$	$^{15}\text{N}-^{15}\text{N}$
A	0	75	25
B	25	75	0
C	50	25	25
D	75	25	0

8 The following statements illustrate the processes that occur during translation, although not necessarily in this order.

- 1 The large subunit of the ribosome binds and forms the translation initiation complex.
- 2 The second amino acyl-tRNA complex now binds to mRNA at the “A” site of the ribosome.
- 3 The small ribosomal subunit, with initiator tRNA bound, binds to the 5’ cap of the mRNA and scans for the first start codon.
- 4 Soluble protein called release factor recognises the stop codon and binds at the “A” site.
- 5 Formation of a peptide bond between the first and the second amino acids by peptidyl transferase.
- 6 The second amino acyl-tRNA complex moves from the “A” site to the “P” site.

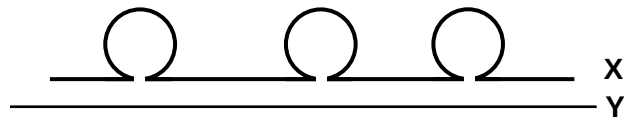
Using the information provided above, deduce the order in which these processes occur.

- A** 1 → 3 → 2 → 5 → 6 → 4
B 1 → 3 → 2 → 6 → 5 → 4
C 3 → 1 → 2 → 5 → 6 → 4
D 3 → 1 → 2 → 6 → 5 → 4

9 Which of the following correctly describes the mutation that gives rise to sickle cell anaemia?

- A** Hydrophilic glutamic acid is changed to hydrophobic valine.
B Hydrophilic valine is changed to hydrophobic glutamic acid.
C Hydrophobic glutamic acid is changed to hydrophilic valine.
D Hydrophobic valine is changed to hydrophilic glutamic acid.

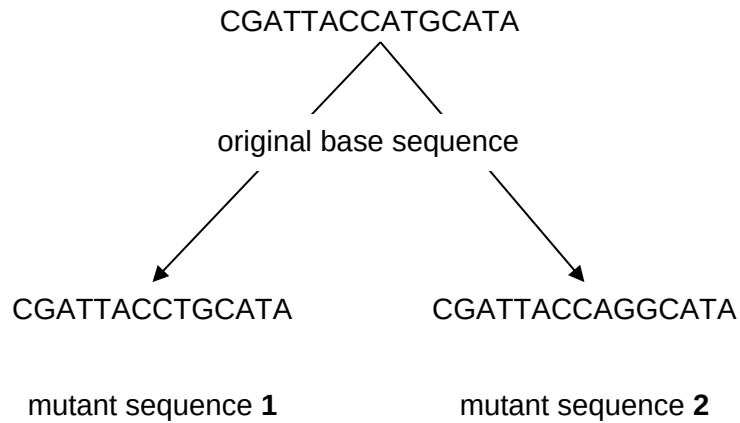
- 10 The diagram below shows the result of hybridizing mature mRNA of ovalbumin to the ovalbumin gene.



Which of the following statement is correct?

	X	Y	Number of exons	Number of introns
A	Mature mRNA	DNA	3	4
B	Mature mRNA	DNA	4	3
C	DNA	Mature mRNA	3	4
D	DNA	Mature mRNA	4	3

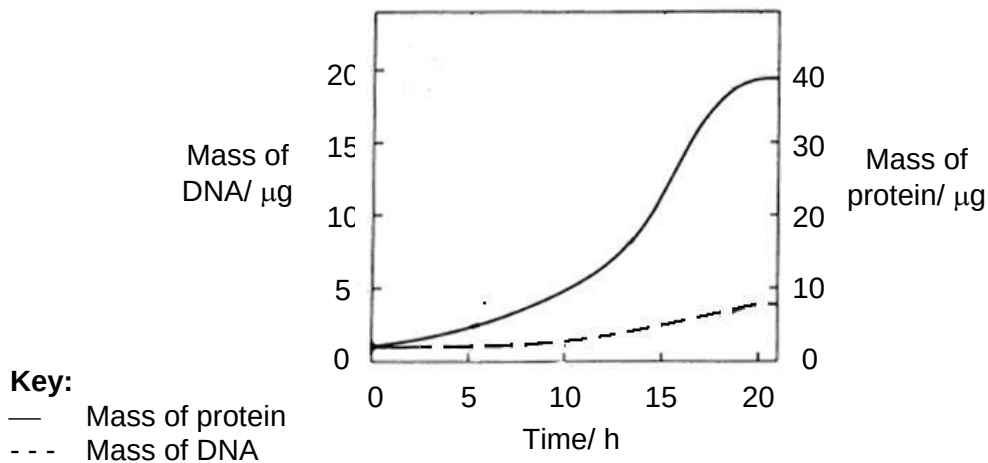
- 11 The diagram below shows the results of two types of gene mutation.



Which one of the following shows the types of gene mutation producing mutant sequences 1 and 2?

	1	2
A	deletion	insertion
B	deletion	substitution
C	substitution	insertion
D	substitution	substitution

- 12** Which of the following statements are true about the end-replication problem of DNA?
- 1 When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
 - 2 Telomerase prevents the end-replication problem from occurring.
 - 3 Repeated rounds of replication produce shorter and shorter DNA molecules.
 - 4 Prokaryotes do not have the end-replication problem.
- A** 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only
- 13** During egg production in *Drosophila*, there is an increased production of eggshell proteins. The genes which code for the eggshell proteins are clustered on the X chromosome.



Which of the following best explains the trends observed in the graph?

- A** Gene amplification has occurred, increasing the rate of transcription. The rate of degradation of poly(A) tail is slower, increasing the duration of translation.
- B** DNA replication has occurred, increasing the rate of and duration of translation.
- C** Gene amplification has occurred, increasing the rate of mutation. The rate of degradation of poly(A) tail is slower, increasing the duration of translation.
- D** DNA replication has occurred. The abundance and activity of eukaryotic translation initiation factors are increased simultaneously, thereby increasing the rate of translation.

14 Four different genes are regulated in different ways.

- Gene 1 undergoes tissue-specific patterns of alternative splicing
- Gene 2 is part of a group of structural genes controlled by the same regulatory sequences
- Gene 3 is in some circumstances subject to methylation
- Gene 4 codes for a repressor protein which acts on an operator site close by

Which row of the table correctly identifies the prokaryotic and eukaryotic genes?

	Prokaryotic	Eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

15 In *E.coli*, repression of the *trp* operon occurs when tryptophan binds to the

- A** operator of *trp* mRNA
- B** operator of DNA
- C** inactive repressor
- D** active repressor

16 How do viruses cause disease in animals?

- 1 They inhibit DNA, RNA or protein synthesis in the host cell.
 - 2 They disrupt the oncogenes of the host cell, thus causing uncontrolled cell division.
 - 3 Their viral proteins and glycoproteins on the surface of host cell cause them to be recognized and destroyed by the body's immune system.
 - 4 They deplete the host cell of cellular materials essential for metabolic functions.
- A** 1 and 4 only
 - B** 3 and 4 only
 - C** 1, 2 and 4 only
 - D** 1, 3 and 4 only

17 Which combination correctly identifies influenza virus?

	RNA genome	DNA genome	Number of strands of viral genome	Protein	Phospholipid
A	✓	✗	2	✓	✓
B	✗	✓	2	✗	✓
C	✓	✗	8	✓	✓
D	✗	✓	8	✓	✗

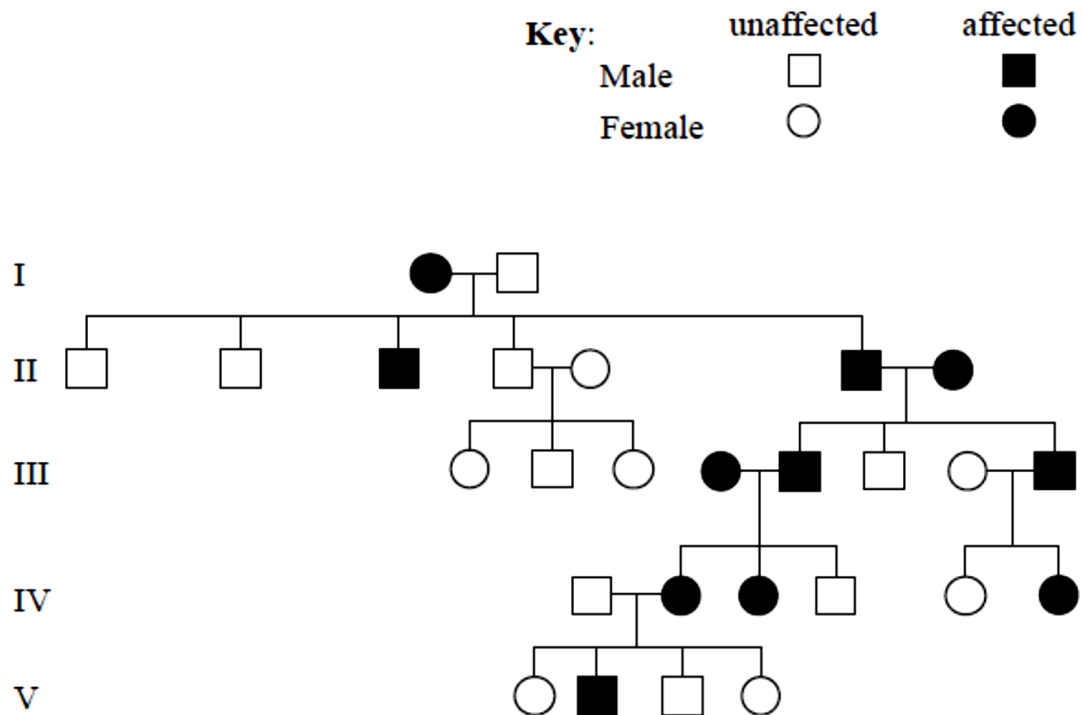
18 In specialized transduction,

- A** any random bacterial gene can be transduced.
- B** only genes on a plasmid can be transduced.
- C** only bacterial genes near the site of integration of the phage DNA can be transduced.
- D** only phage genes near the recipient bacteria integration site can be transduced.

19 Which of the following would cause phenotypic variation among organisms of the same genotype?

- A** Mutation
- B** Continuous variation within the species
- C** Exposure to different environments
- D** Different varieties of the same species

20 The pedigree chart below shows the inheritance of a genetic disease in a family.



What is the nature of the allele that causes this disease?

- A Dominant and sex linked
- B Dominant and non-sex linked
- C Recessive and sex linked
- D Recessive and non-sex linked

21 Which one of the following features is an example of discontinuous variation in humans?

- A blood group
- B body mass
- C height
- D skin colour

22 A linkage group is composed of genes which

- A** associate during prophase I.
- B** are in the same nucleus.
- C** segregate at mitosis.
- D** are situated on the same chromosome.

23 A pure-breeding plant with the dominant phenotype of character **P** and the recessive phenotype of character **Q** was crossed with another pure-breeding plant with the recessive phenotype of character **P** and the dominant phenotype of **Q**. The offspring of this cross were crossed with a double homozygous recessive for **P** and **Q** and the following results were obtained:

22 were phenotypically dominant for **P** and recessive for **Q**.

5 were phenotypically dominant for both **P** and **Q**.

4 were phenotypically recessive for both **P** and **Q**.

24 were phenotypically recessive for **P** and dominant for **Q**.

Which one of the following types of inheritance is illustrated by these results?

- A** gene linkage of **P** and **Q**
- B** Mendelian dihybrid inheritance
- C** multiple alleles
- D** polygenic inheritance

- 24** The α , β and γ globin chains of human, chimpanzee, gorilla and gibbon haemoglobin were analysed. The number of differences in the amino acid sequences in comparison with the human molecules is shown in the table.

species	Number of differences in amino acid sequence from human molecules		
	α globin	β globin	γ globin
chimpanzee	0	0	1
gorilla	1	1	1
gibbon	3	3	2

What do the differences suggest?

- A** Humans and gorillas are more closely related than humans and gibbons.
 - B** Humans and gibbons do not share a common ancestor.
 - C** Gibbons are the ancestor of gorillas and chimpanzees are the ancestor of humans.
 - D** Humans and gibbons are more closely related than humans and chimpanzees.
- 25** A large lake was inhabited by one species of fish. Over time it is inhabited by many different species of fish, each of them is a descendant of the original population which inhabited the large lake.

This is an example of

- A** artificial selection
- B** natural selection
- C** allopatric speciation
- D** sympatric speciation

26 Which of the following are regarded as evidence for evolution?

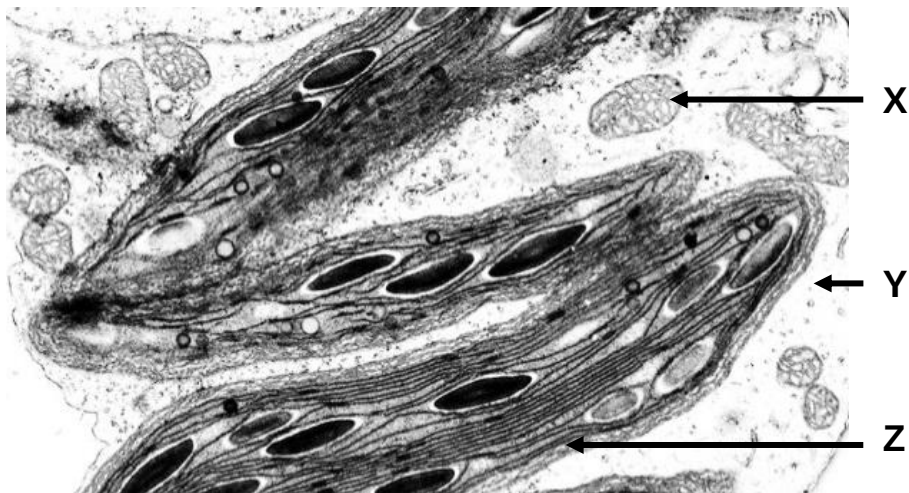
- 1 Embryological homology
- 2 Fossil record
- 3 Biogeography
- 4 Analogous structures

- A 1, 3 only
 B 2, 3 only
 C 1, 2, 3 only
 D All of the above

27 How many molecules of NADH and FADH_2 will be produced given **two** molecules of pyruvate?

- A 3 NADH, 1 FADH_2
 B 4 NADH, 1 FADH_2
 C 6 NADH, 2 FADH_2
 D 8 NADH, 2 FADH_2

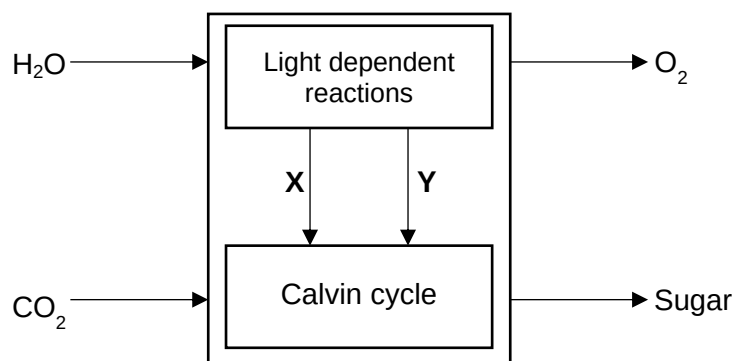
28 The diagram shows the ultrastructure of a plant cell.



Which of the following row is correct?

	Site of ATP synthesis	Site of carbohydrate synthesis
A	X	Z
B	Z	X
C	X, Y	X
D	X, Y, Z	Z

- 29 The diagram below shows the processes that occur in the chloroplast during photosynthesis.



What are X and Y?

	X	Y
A	ATP	NADH
B	ATP	NADPH
C	Protons	Electrons
D	FADH ₂	NADH

- 30 What is a similarity between G-protein-linked receptors and tyrosine-kinase receptors?
- A Both are activated upon dimerization.
 - B Both have hydrophobic stretches of amino acids that serve to anchor the receptors in the membrane.
 - C Both have intracellular domains that are involved in ligand binding.
 - D Both rely solely on protein kinase activity to amplify the signal received.
- 31 Impulses travel very rapidly along nerves from the leg muscles of a mammal because
- A there is a high concentration of Na⁺ ions inside the axons.
 - B the nerves contain myelinated fibres.
 - C there is a potential difference across the axon membranes
 - D a nerve impulse is an all-or-nothing phenomenon.

- 32** Viral RNA polymerase is studied using genetic engineering techniques. Some of these stages are listed.

- 1 cloning into an expression vector
- 2 disruption of cells and isolation of cytoplasmic fraction
- 3 expression of viral protein
- 4 selection of the bacteria with recombinant plasmid
- 5 PCR (polymerase chain reaction) of viral cDNA
- 6 reverse transcription of viral genomic RNA
- 7 isolation of viral genomic RNA from virus
- 8 transformation into *Escherichia coli* cells

What is the correct order?

- A** 7, 2, 3, 1, 8, 5, 6, 4
 - B** 7, 6, 5, 1, 8, 4, 3, 2
 - C** 7, 6, 5, 8, 4, 2, 3, 1
 - D** 8, 4, 1, 2, 7, 6, 3, 5
- 33** *lacZ* gene is a genetic marker found in the plasmid which can be used in genetic engineering.

What is the function of *lacZ* in a cloning vector?

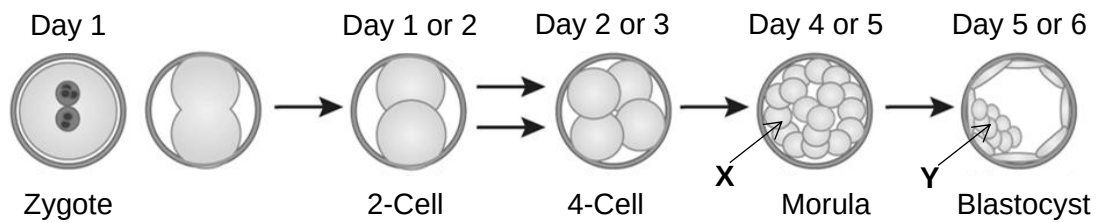
- A** Express *lac* repressor
- B** Distinguish between introns and exons
- C** Break down lactose to galactose and glucose
- D** Screen for cells with the recombinant plasmid

34 Which of the following statements are true about adult stem cells?

- 1 They can undergo many rounds of mitosis.
- 2 They can be totipotent, pluripotent or multipotent.
- 3 They can differentiate into almost any cell type.
- 4 They can give rise to specialised cells.

- A** 1 and 4 only
B 3 and 4 only
C 1, 2 and 4 only
D 1, 3 and 4 only

35 The figure below shows several stages in the development of an embryo.



Which of the following statement is true about the cells labelled **X** and **Y**?

- A** **X** is a pluripotent cell while **Y** is a multipotent cell.
B **X** is a totipotent cell while **Y** can divide to give multipotent cells.
C Both **X** and **Y** are totipotent.
D **X** can divide to give totipotent cells while **Y** can divide to give pluripotent cells.

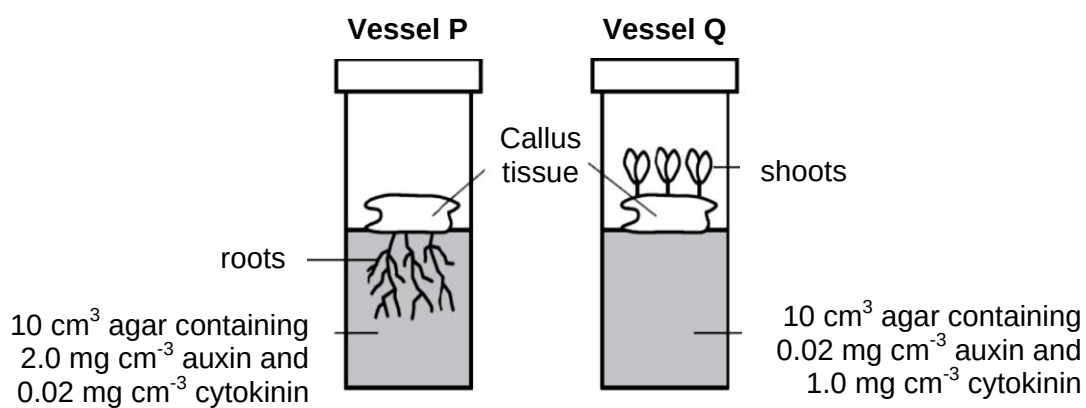
- 36 Which of the following has the correct information about the genetic diseases and their corresponding forms of gene therapy?

	Genetic Disease	Inheritance Pattern	Gene Therapy
A	Cystic Fibrosis	Sex-linked recessive	Recombinant DNA molecule is packaged in a modified adenovirus
B	Cystic Fibrosis	Autosomal recessive	Recombinant RNA molecule is packaged in a modified adenovirus
C	ADA-SCID	Autosomal recessive	Recombinant RNA molecule is packaged in a modified retrovirus
D	X-linked SCID	Sex-linked recessive	Recombinant DNA molecule is packaged in a modified retrovirus

- 37 Which row correctly states the effects and possible consequences of producing the various genetically-modified organisms?

	Bt corn		Golden rice		GM salmon	
	<i>effect</i>	<i>possible consequence</i>	<i>effect</i>	<i>possible consequence</i>	<i>effect</i>	<i>possible consequence</i>
A	improve quality	become weeds	improve yield	become weeds	improve yield	displace native species
B	improve yield	become weeds	improve quality	cause allergies	improve yield	displace native species
C	improve yield	become weeds	improve yield	cause allergies	improve quality	cause allergies
D	improve yield	become weeds	improve quality	become weeds	improve yield	cause allergies

38 The figure shows the results of an experiment involving callus culture.

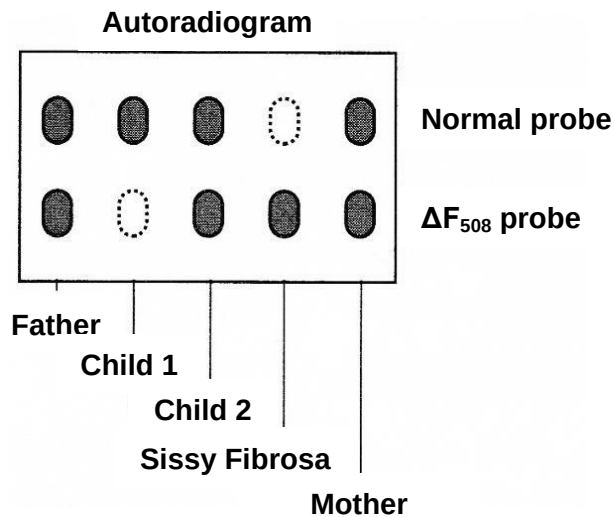


Which of the following statements explains the result of the experiment?

- A A high auxin to cytokinin ratio promotes root while a low auxin to cytokinin ratio promotes shoot formation.
- B The cells making up the callus are uniform with no variations in morphology and physiology.
- C The callus consists of actively dividing cells while the roots and shoots contain differentiated cells.
- D High auxin level inhibits shoot formation while high cytokinin level inhibits root formation.

- 39** A method that uses allele-specific oligonucleotide probes has been developed to detect cystic fibrosis. Oligonucleotide probes, complementary to the region where a 3-base deletion is located, have been chemically synthesised. One probe binds to the mutant (ΔF_{508}) allele whilst the other binds to the normal allele.

DNA samples from Sissy Fibrosa, her parents and two siblings were isolated and amplified. Their DNA samples were blotted on nitrocellulose paper and treated with oligonucleotide probes. The results obtained were shown in the figure below. Dark spots indicate binding of probe.



Which members of Sissy's family have cystic fibrosis, which are normal and which are carriers?

	cystic fibrosis	normal	carriers
A	Child 1	Sissy Fibrosa	Father, Mother, Child 2
B	Child 1	Father, Mother, Child 2	Sissy Fibrosa
C	Sissy Fibrosa	Father, Mother, Child 2	Child 1
D	Sissy Fibrosa	Child 1	Father, Mother, Child 2

40 Some of the goals and benefits of Human Genome Project (HGP) are listed below.

- 1 Prescribe the right drug, in the right dosage and combination to each individual based on the genotype of the individual
- 2 Develop computer programs or software to analyze the data from HGP
- 3 Design drugs based on the primary structure and the 3D shape of the target proteins
- 4 Develop genetic tests to screen for carriers of genetic disorders
- 5 Develop more efficient methods for DNA sequencing and sequence analysis
- 6 Create a genetic linkage map and physical map of the human genome

Which row correctly identifies the goals and benefits of HGP?

	Goals	Benefits
A	1, 3, 4	2, 5, 6
B	1, 4, 5	2, 3, 6
C	2, 3, 6	1, 4, 5
D	2, 5, 6	1, 3, 4

TJC Prelim 2015
H2 Biology Paper 1 (9648/01) Answers

Question	Answer	Question	Answer
1	D	21	A
2	A	22	D
3	B	23	A
4	C	24	A
5	A	25	D
6	B	26	D
7	D	27	D
8	C	28	D
9	A	29	B
10	D	30	B
11	B	31	B
12	D	32	B
13	A	33	D
14	D	34	A
15	C	35	D
16	D	36	C
17	C	37	B
18	C	38	A
19	C	39	D
20	B	40	D